Nishmitha Devadiga

Bangalore, India | +91 6366040691 | nishmithadev2005@gmail.com | linkedin.com/in/nishmitha-devadiga | github.com/nishmithadev | nishmithadev.github.io

Education

SJB Institute of Technology — Bangalore, India
B.E. in Computer Science and Engineering (Data Science Specialization); CGPA: 9.08/10 — *Expected April 2027*

Projects

Vachaka – AI Sign Language Communication Platform | *React, Tailwind CSS, FastAPI, TensorFlow, MediaPipe*

- Built a full-stack web application that translates between American Sign Language (ASL) and spoken language in real time, enabling two-way communication for the deaf and hard-of-hearing community
- Trained a TensorFlow deep learning model on 87,000+ ASL images, achieving **98.67% recognition accuracy**, and integrated MediaPipe for real-time hand landmark extraction from webcam input
- Engineered a Speech-to-Sign pipeline converting spoken words/sentences into corresponding ASL gestures, alongside letter-spelling and quick-phrase modes for natural conversation flow
- Developed the frontend in React with Tailwind CSS and a FastAPI backend to serve real-time inference with low latency

Risk-Aware Multi-Stock Portfolio Optimization using Deep RL | *Python, TensorFlow, OpenAI Gym, NLP, Streamlit*

- Designed an autonomous portfolio management system combining Deep Reinforcement Learning (DQN) with NLP-based sentiment analysis to make Buy/Sell/Hold decisions across multiple stocks simultaneously
- Built a custom OpenAI Gym-style trading environment using OHLCV data, technical indicators (RSI, MACD, Moving Averages), and financial news sentiment scores as state inputs
- Implemented a Deep Q-Network agent using the Bellman equation, experience replay, and ϵ-greedy exploration, optimized via a risk-adjusted reward function based on the Sharpe Ratio rather than raw profit
- Benchmarked agent performance against Buy-and-Hold and random baselines using Cumulative Return, Sharpe Ratio, Max Drawdown, and Volatility; visualized results in an interactive Streamlit and Plotly dashboard

HemoSmart – Intelligent Blood Management Platform | *XGBoost, LSTM/Prophet, LangChain, RAG, MCP*

- Led the design of an agentic, multi-agent AI platform (college major project) to shift hospital blood management from reactive tracking to proactive, predictive resource planning
- Built ML models (XGBoost, Random Forest) to predict individual patient transfusion needs from clinical parameters, and LSTM/Prophet time-series models to forecast hospital-wide weekly blood demand
- Integrated a Retrieval-Augmented Generation (RAG) explanation engine grounded in medical guidelines to make every AI-driven recommendation clinically transparent and trustworthy
- Orchestrated model coordination via a multi-agent architecture (LangChain/CrewAI) and connected the platform to hospital inventory systems using the Model Context Protocol (MCP) to reduce blood wastage and shortages

Technical Skills

Languages: Python, Java
Web & APIs: React, Tailwind CSS, FastAPI
AI/ML & Data: TensorFlow, MediaPipe, XGBoost, LSTM/Prophet, Reinforcement Learning (DQN), NLP, LangChain/CrewAI, RAG
Tools & Concepts: Git/GitHub, DBMS, Computer Networking, Cloud Computing, Streamlit, Plotly
Relevant Coursework: Database Management Systems, Computer Networking, Cloud Computing, Machine Learning, Artificial Intelligence

Certifications

NPTEL: **Responsible and Safe AI Systems**
NPTEL: **Introduction to Database Systems**

Extracurricular Activities

Member, College Fashion Team – Won awards at the Intercollege Fest 2025

Languages

English, Kannada, Hindi (Fluent) | Japanese (Currently Learning)